

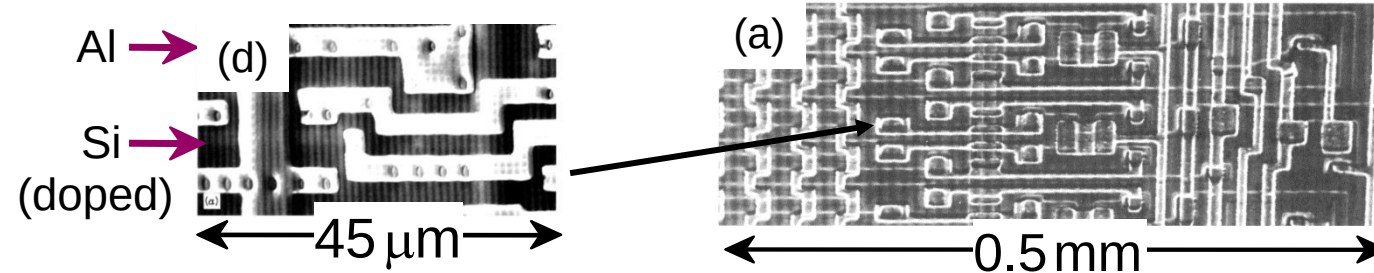
Chapter 18: Electrical Properties

ISSUES TO ADDRESS...

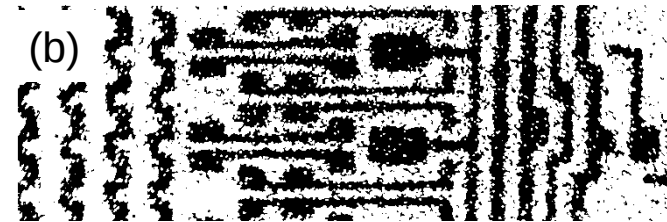
- How are electrical conductance and resistance characterized?
- What are the physical phenomena that distinguish conductors, semiconductors, and insulators?
- For metals, how is conductivity affected by imperfections, temperature, and deformation?
- For semiconductors, how is conductivity affected by impurities (doping) and temperature?

View of an Integrated Circuit

- Scanning electron micrographs of an IC:



- A dot map showing location of Si (a semiconductor):
 - Si shows up as light regions.



- A dot map showing location of Al (a conductor):
 - Al shows up as light regions.

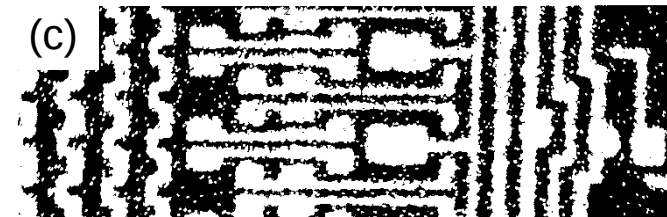


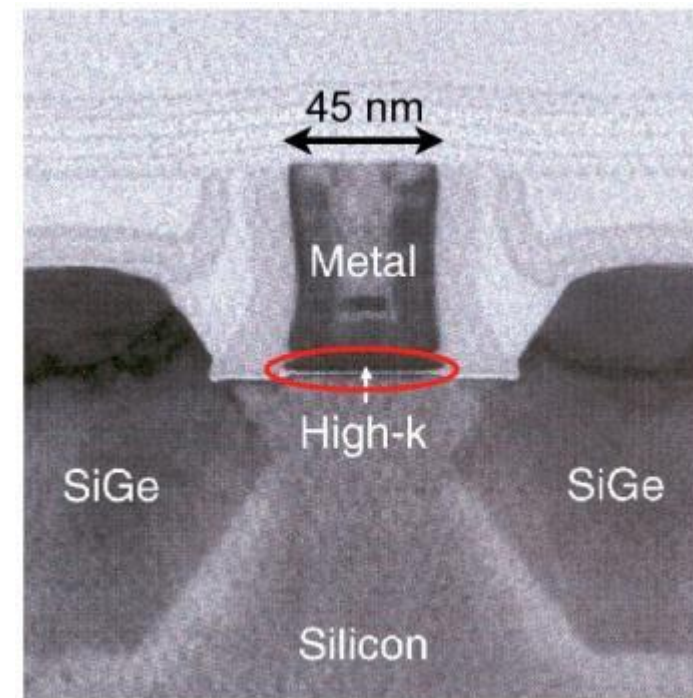
Fig. (d) from Fig. 12.27(a), *Callister & Rethwisch 3e*.
(Fig. 12.27 is courtesy Nick Gonzales, National Semiconductor Corp., West Jordan, UT.)

Figs. (a), (b), (c) from Fig. 18.27, *Callister & Rethwisch 8e*.

Example – MOSFET

- MOSFET
 - Metal-Oxide-Semiconductor
 - Field-effect transistor
 - A simple switch
 - ~ 12 billion FET in each state-of-the-art transistor
 - Electron mobility

TEM image of a single transistor from Intel's 45nm technology

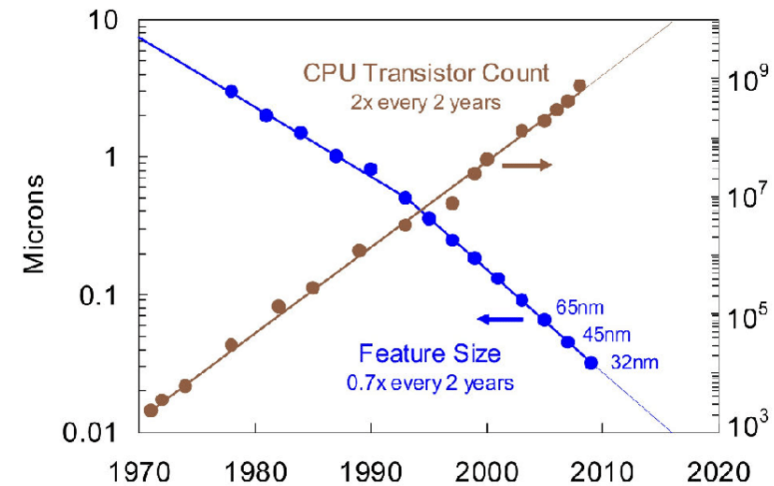


Adapted from:
<https://volga.eng.yale.edu/teaching-resources/semiconductors>

Example – MOSFET

- MOSFET
 - Metal-Oxide-Semiconductor
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Moore's Law/Observation



Adapted from:

<https://www.semanticscholar.org/paper/Moore's-Law-Past-32nm%3A-Future-Challenges-in-Device-Kuhn/177319f349b78259b79c3e76dc308502544cffc5>

Electrical Conduction

- Ohm's Law:

voltage drop (volts = J/C)
C = Coulomb

$$V = IR$$

current (amps = C/s)

resistance (Ohms)

- Resistivity, ρ :

-- a material property that is independent of sample size and geometry

$$\rho = \frac{RA}{l}$$

surface area of current flow

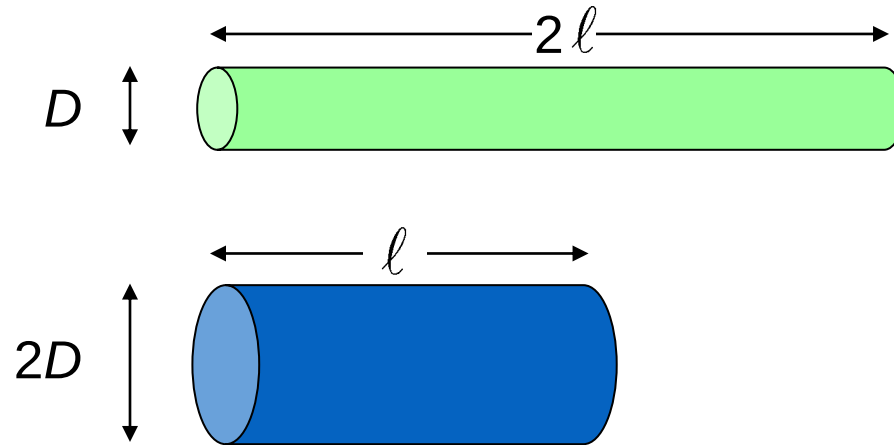
current flow path length

- Conductivity, σ

$$\sigma = \frac{1}{\rho}$$

Electrical Properties

- Which will have the greater resistance?



$$R_1 = \frac{2\rho l}{\pi \left(\frac{D}{2}\right)^2} = \frac{8\rho l}{\pi D^2}$$

$$R_2 = \frac{\rho l}{\pi \left(\frac{2D}{2}\right)^2} = \frac{\rho l}{\pi D^2} = \frac{R_1}{8}$$

- Analogous to flow of water in a pipe
- Resistance depends on sample geometry and size.

Definitions

Further definitions

$$\boxed{J = \sigma \varepsilon} \quad \Leftarrow \text{another way to state Ohm's law}$$

$$J \equiv \text{current density} = \frac{\text{current}}{\text{surface area}} = \frac{I}{A} \quad \text{like a flux}$$

$$\varepsilon \equiv \text{electric field potential} = V/\ell$$

$$J = \sigma (V/\ell)$$

Diagram illustrating the components of the equation $J = \sigma (V/\ell)$:

- J is labeled as **Electron flux** (orange).
- σ is labeled as **conductivity** (red).
- V/ℓ is labeled as **voltage gradient** (green).

Conductivity: Comparison

- Room temperature values $(\text{Ohm-m})^{-1} = (\Omega \cdot \text{m})^{-1}$

METALS

conductors

Silver	6.8×10^7
Copper	6.0×10^7
Iron	1.0×10^7

CERAMICS

Soda-lime glass	10^{-10} - 10^{-11}
Concrete	10^{-9}
Aluminum oxide	$<10^{-13}$

SEMICONDUCTORS

Silicon	4×10^{-4}
Germanium	2×10^0
GaAs	10^{-6}

semiconductors

POLYMERS

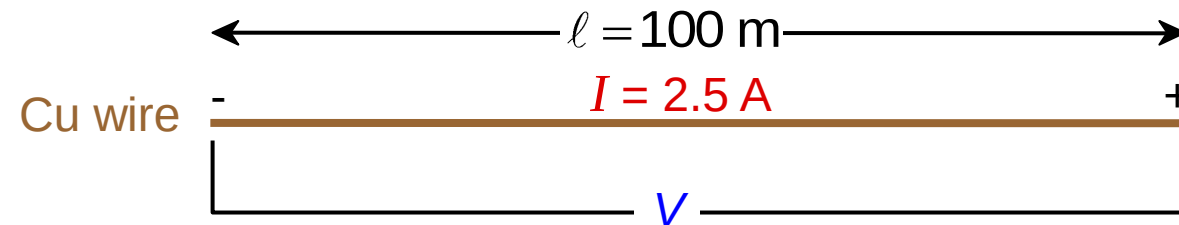
Polystyrene	$<10^{-14}$
Polyethylene	10^{-15} - 10^{-17}

insulators

Selected values from Tables 18.1, 18.3, and 18.4, *Callister & Rethwisch 8e*.

Example: Conductivity Problem

What is the minimum diameter (D) of the wire so that $V < 1.5$ V?



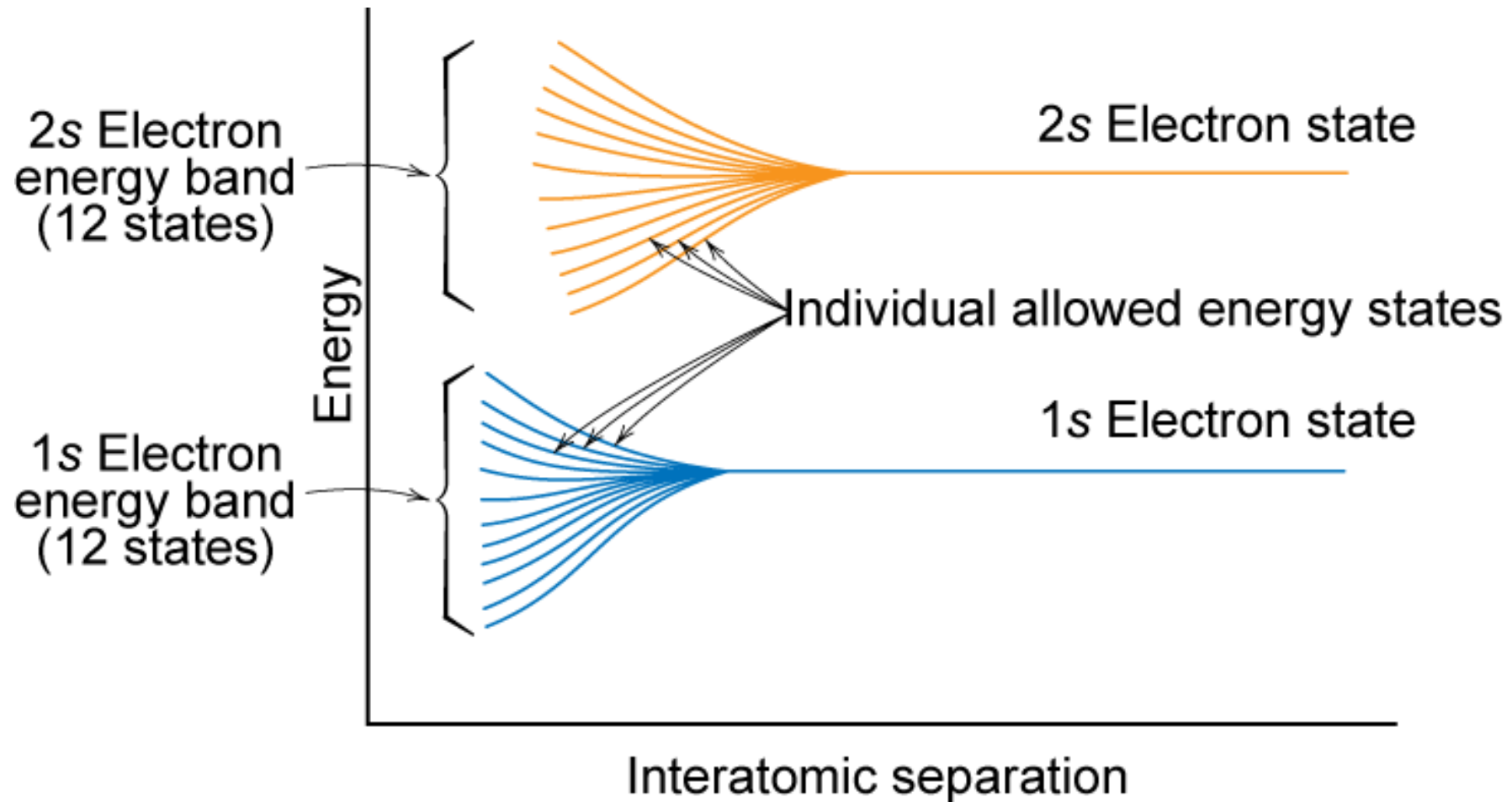
$$R = \frac{\ell}{A\sigma} = \frac{V}{I}$$

Diagram illustrating the formula for resistance R of a wire:

- ℓ (length) is labeled as 100 m.
- A (cross-sectional area) is labeled as $\frac{\pi D^2}{4}$.
- σ (conductivity) is labeled as $6.07 \times 10^7 \text{ (Ohm-m)}^{-1}$.
- V (voltage) is labeled as < 1.5 V.
- I (current) is labeled as 2.5 A.

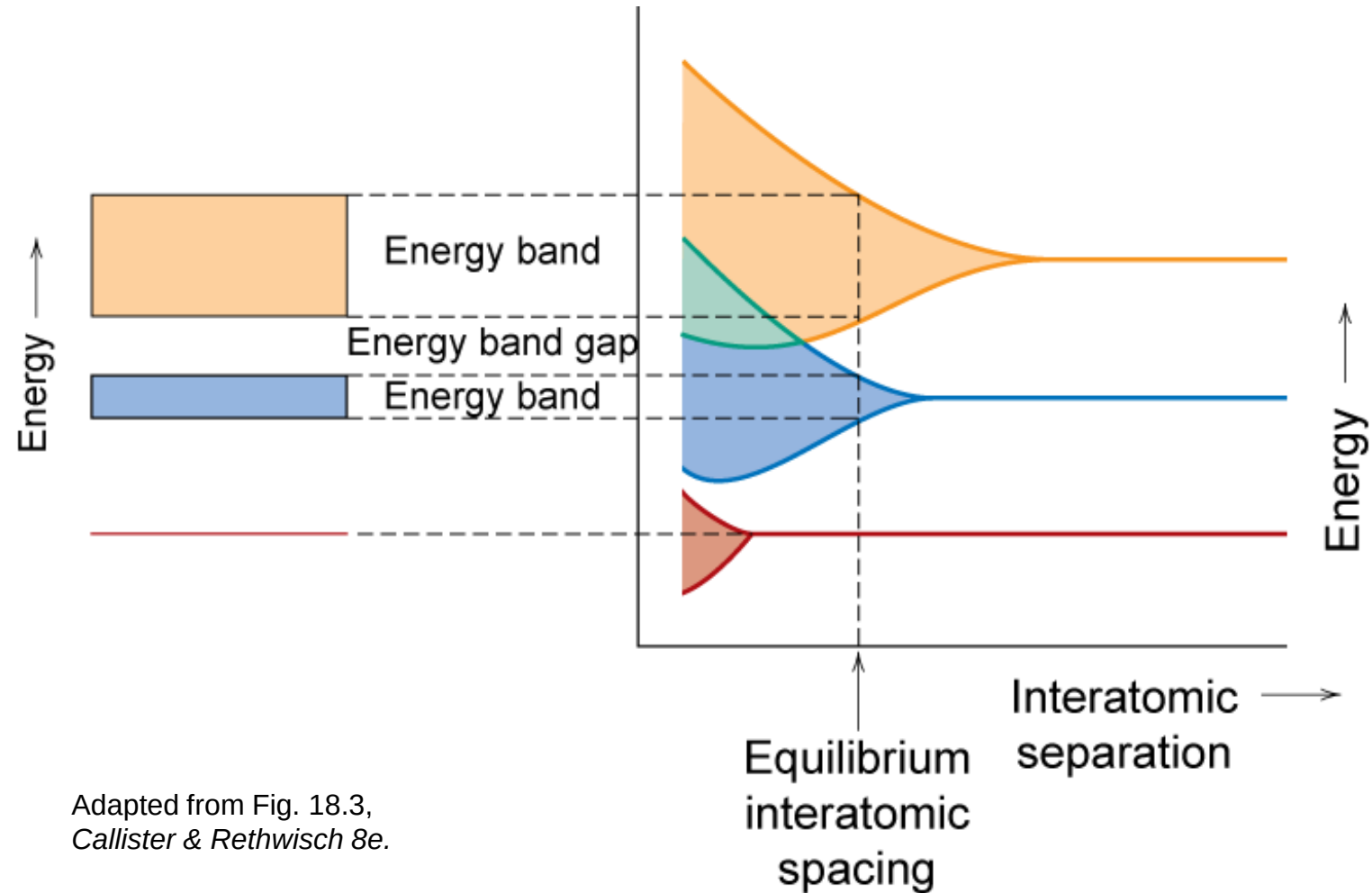
Solve to get $D > 1.87$ mm

Electron Energy Band Structures

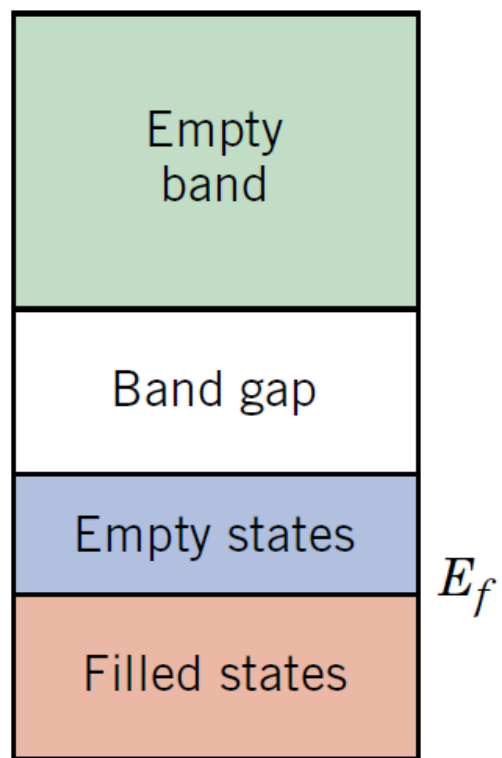


Adapted from Fig. 18.2, *Callister & Rethwisch 8e*.

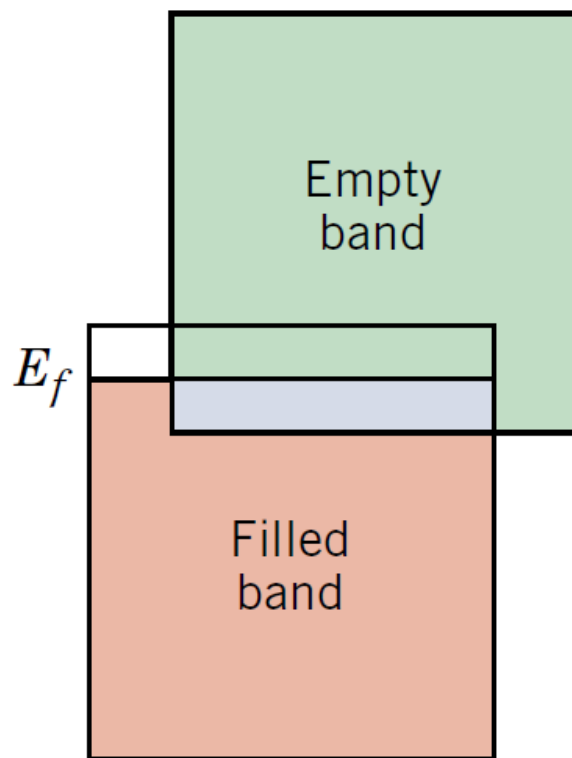
Band Structure Representation



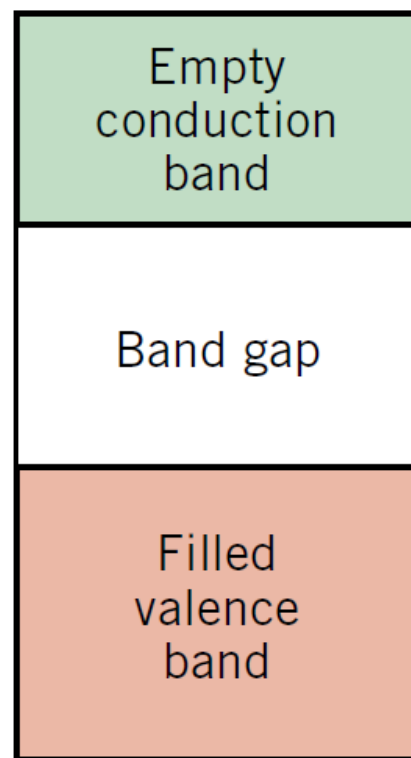
Adapted from Fig. 18.3,
Callister & Rethwisch 8e.



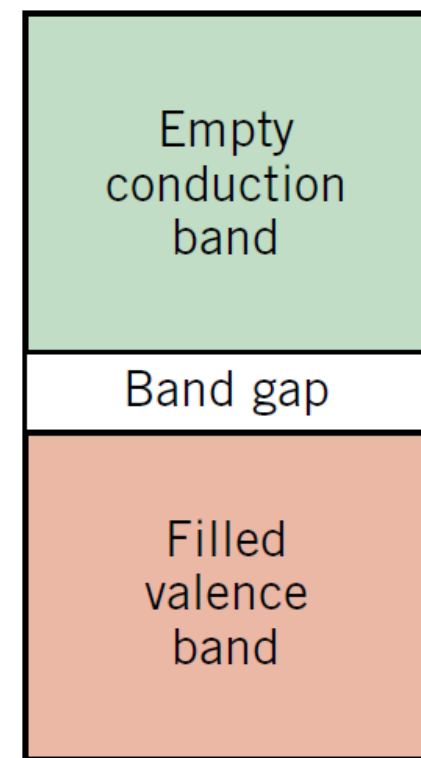
(a)



(b)



(c)



(d)

Conduction & Electron Transport

- Metals (**Conductors**):

- for metals empty energy states are adjacent to filled states.

- thermal energy excites electrons into empty higher energy states.

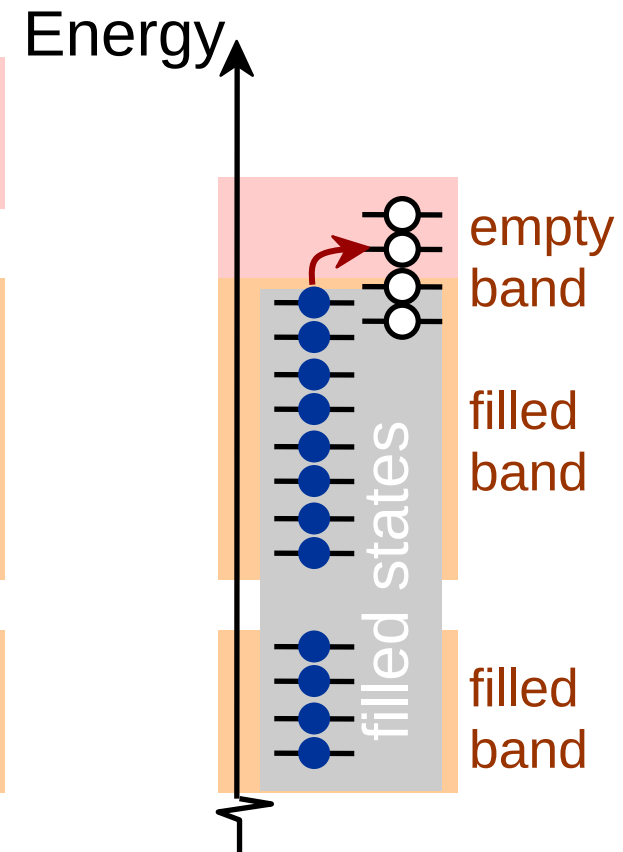
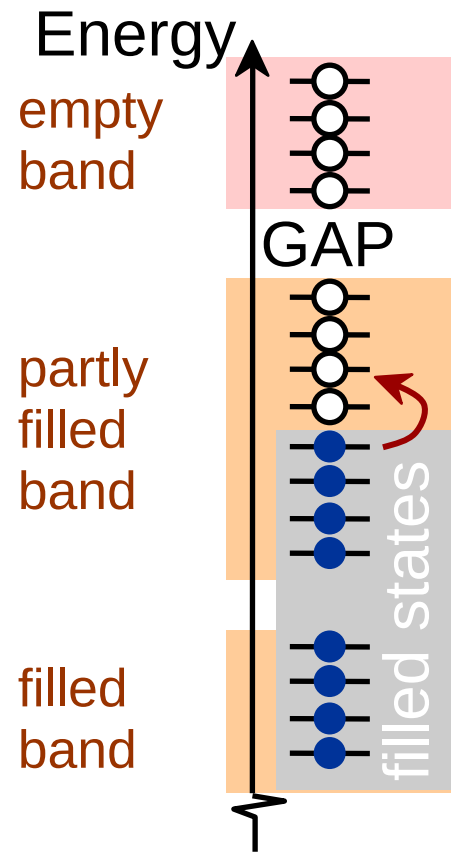
- two types of band structures for metals

- partially filled band

- empty band that overlaps filled band

Partially filled band

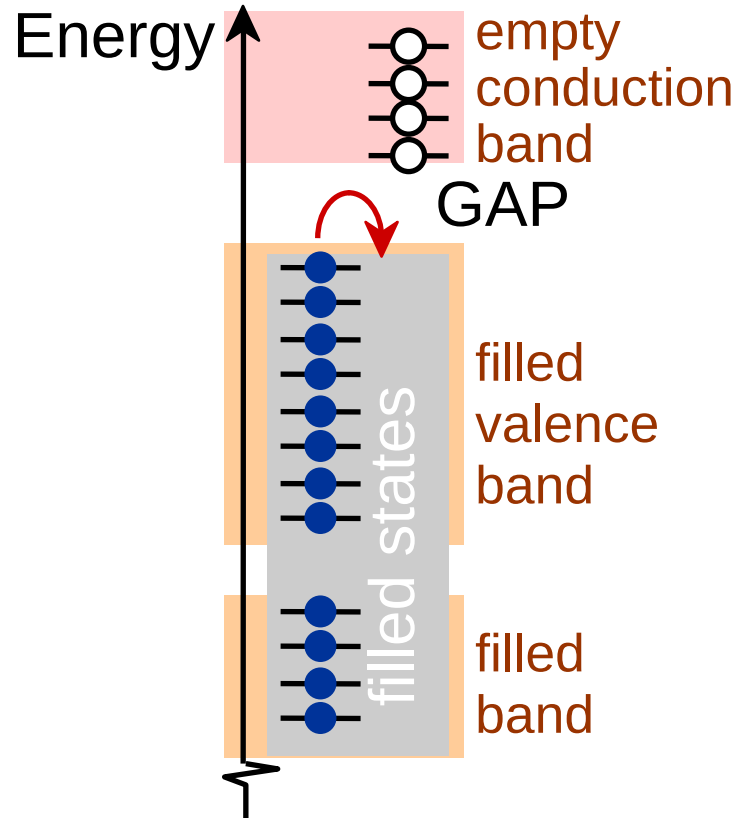
Overlapping bands



Energy Band Structures: Insulators & Semiconductors

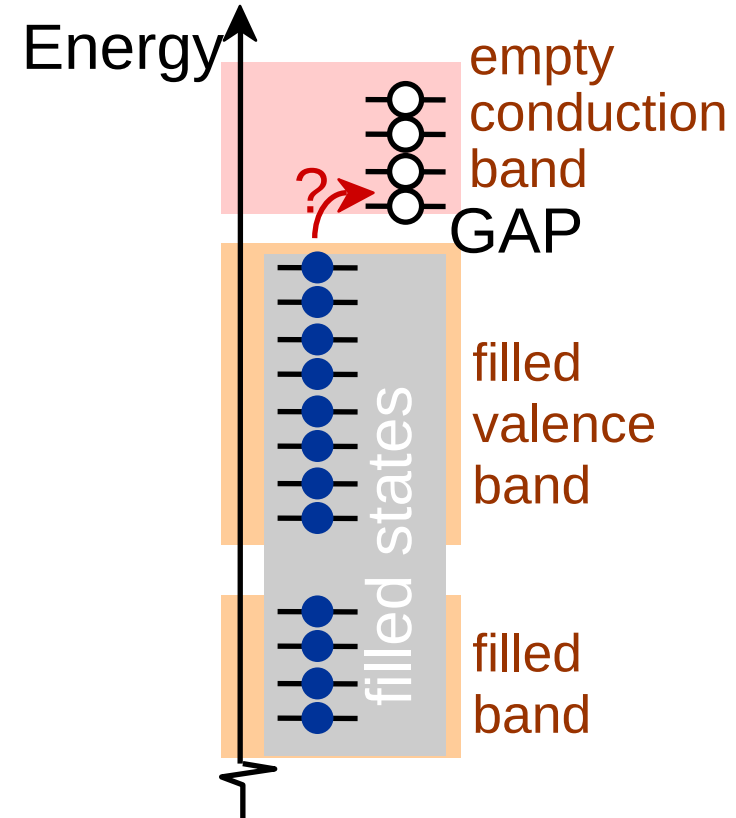
- Insulators:

- wide band gap (> 2 eV)
- few electrons excited across band gap



- Semiconductors:

- narrow band gap (< 2 eV)
- more electrons excited across band gap

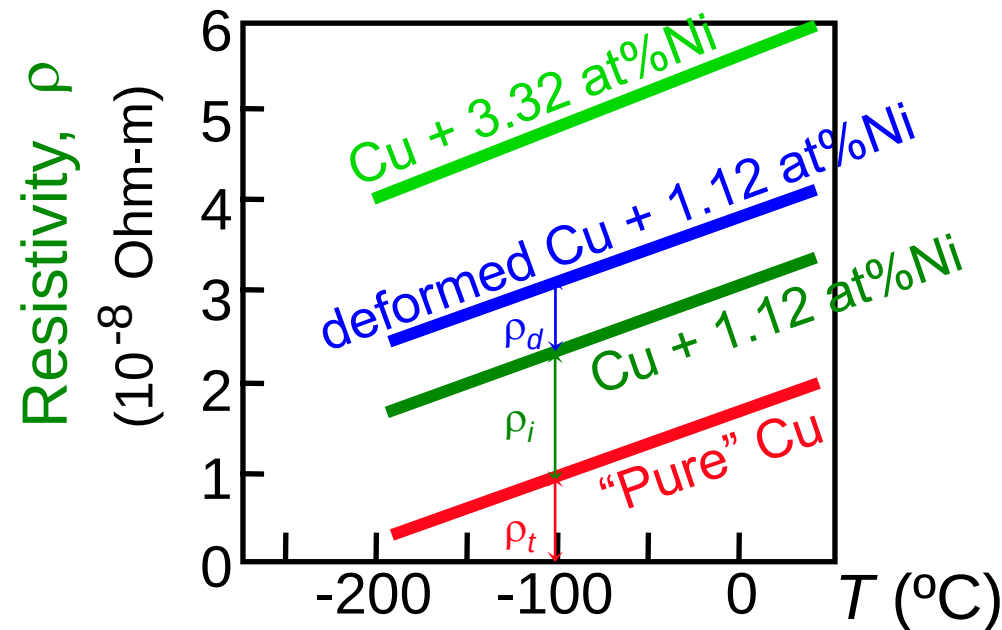


Metals: Influence of Temperature and Impurities on Resistivity

- Presence of imperfections increases resistivity

- grain boundaries
- dislocations
- impurity atoms
- vacancies

These act to scatter electrons so that they take a less direct path.



- Resistivity

increases with:

- temperature
- wt% impurity
- %CW

$$\rho = \rho_{\text{thermal}} + \rho_{\text{impurity}} + \rho_{\text{deformation}}$$

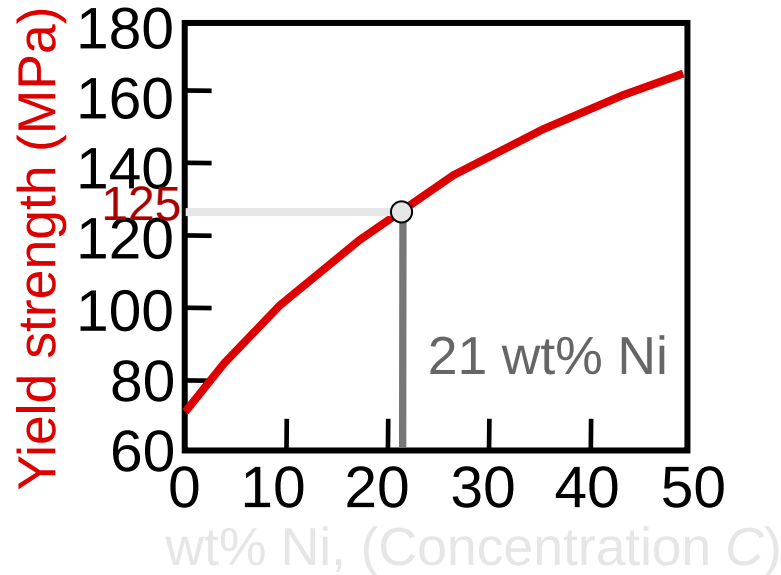
Adapted from Fig. 18.8, Callister & Rethwisch 8e. (Fig. 18.8 adapted from J.O. Linde, *Ann. Physik* **5**, p. 219 (1932); and C.A. Wert and R.M. Thomson, *Physics of Solids*, 2nd ed., McGraw-Hill Book Company, New York, 1970.)

Estimating Conductivity

- Question:

- Estimate the electrical conductivity σ of a Cu-Ni alloy that has a yield strength of **125 MPa**.

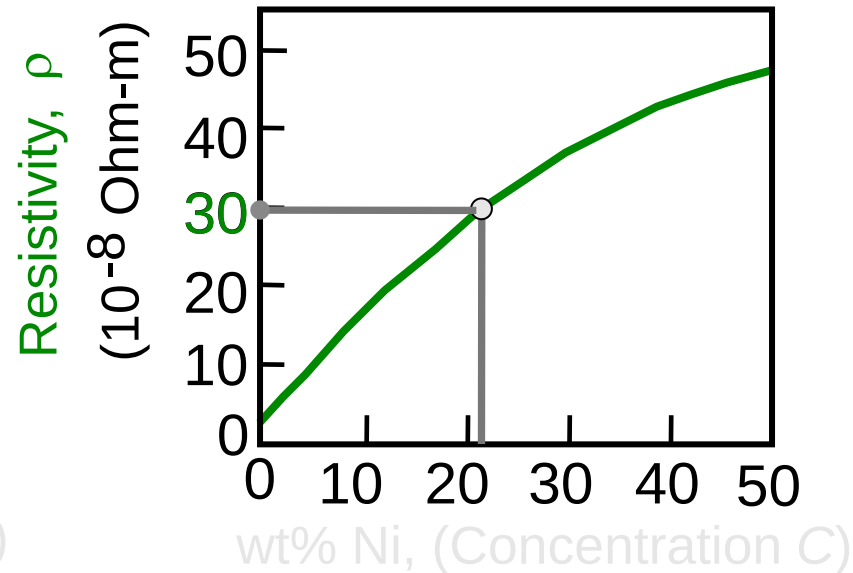
Adapted from Fig. 18.9, Callister & Rethwisch 8e.



Adapted from Fig. 7.16(b), Callister & Rethwisch 8e.

From step 1:

$$C_{\text{Ni}} = 21 \text{ wt\% Ni}$$

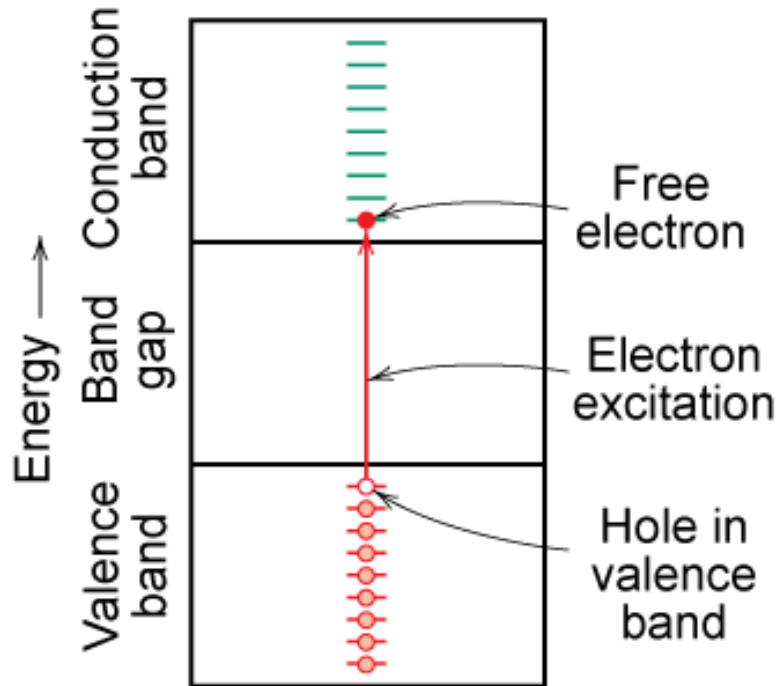


$$\rho = 30 \times 10^{-8} \text{ Ohm} - \text{m}$$

$$\sigma = \frac{1}{\rho} = 3.3 \times 10^6 (\text{Ohm} - \text{m})^{-1}$$

Charge Carriers in Insulators and Semiconductors

Adapted from Fig. 18.6(b),
Callister & Rethwisch 8e.



Two types of electronic charge carriers:

Free Electron

- negative charge
- in conduction band

Hole

- positive charge
- vacant electron state in the valence band

Move at different speeds - **drift velocities**

Intrinsic Semiconductors

- Pure material semiconductors: e.g., silicon & germanium
 - Group IVA materials
- Compound semiconductors
 - III-V compounds
 - Ex: GaAs & InSb
 - II-VI compounds
 - Ex: CdS & ZnTe
 - The wider the electronegativity difference between the elements the wider the energy gap.

Drude Model

$$\bar{F} \propto -e \bar{E}$$

$\underbrace{1.6 \times 10^{-19}}_{\text{m}} [\text{C}]$

$$e \quad \text{---} \quad dx$$

l : mean free path

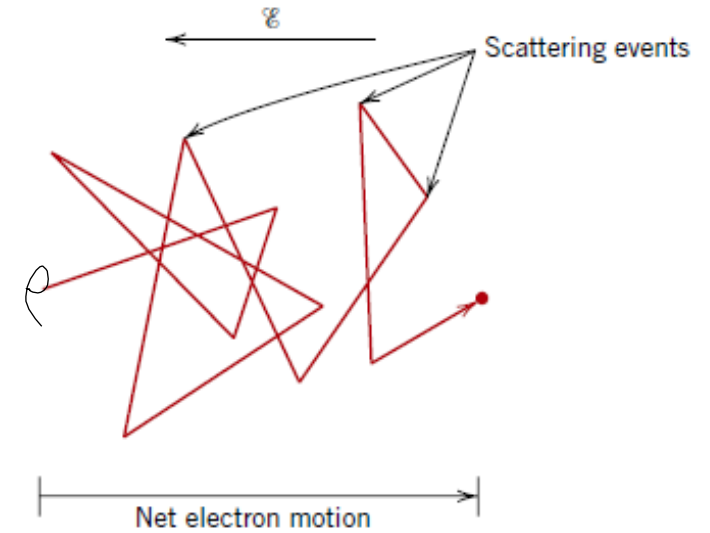
τ : mean free time

$$v_d \propto \frac{l}{\tau}$$

$$\frac{v_d}{E} = \mu \left[\frac{\text{m}^2}{\text{V.s}} \right]$$

$$J = n \mu e E$$

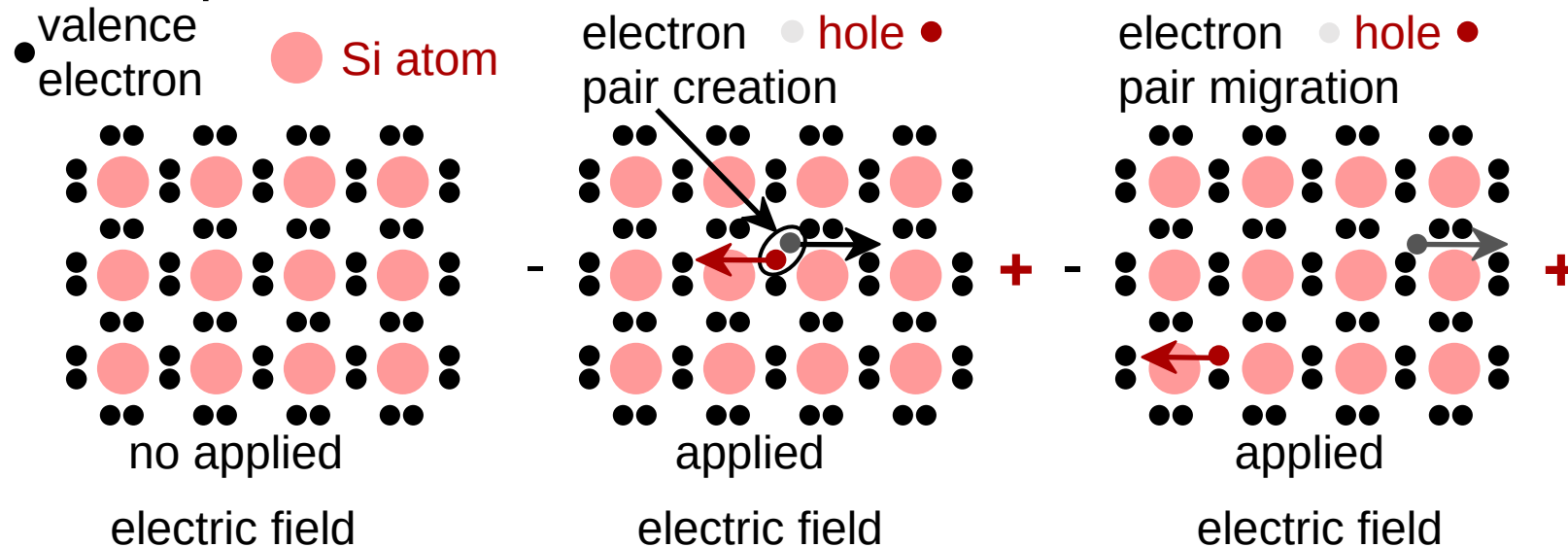
$$\Rightarrow j = (n \mu e) E$$



<i>Material</i>	<i>Band Gap (eV)</i>	<i>Electrical Conductivity [($\Omega\cdot m$)⁻¹]</i>	<i>Electron Mobility (m²/V·s)</i>	<i>Hole Mobility (m²/V·s)</i>
Elemental				
Si	1.11	4×10^{-4}	0.14	0.05
Ge	0.67	2.2	0.38	0.18
III–V Compounds				
GaP	2.25	—	0.03	0.015
GaAs	1.42	10^{-6}	0.85	0.04
InSb	0.17	2×10^4	7.7	0.07
II–VI Compounds				
CdS	2.40	—	0.03	—
ZnTe	2.26	—	0.03	0.01

Intrinsic Semiconduction in Terms of Electron and Hole Migration

- Concept of electrons and holes:



Adapted from Fig. 18.11,
Callister & Rethwisch 8e.

- Electrical Conductivity given by:

$$\sigma = n|e|\mu_e + p|e|\mu_h$$

electrons/m³ → n

electron mobility → μ_e

holes/m³ → p

hole mobility → μ_h

Number of Charge Carriers

Intrinsic Conductivity

$$\sigma = n|e|\mu_e + p|e|\mu_h$$

- for intrinsic semiconductor $n = p = n_i$

$$\therefore \sigma = n_i|e|(\mu_e + \mu_h)$$

- Ex: GaAs

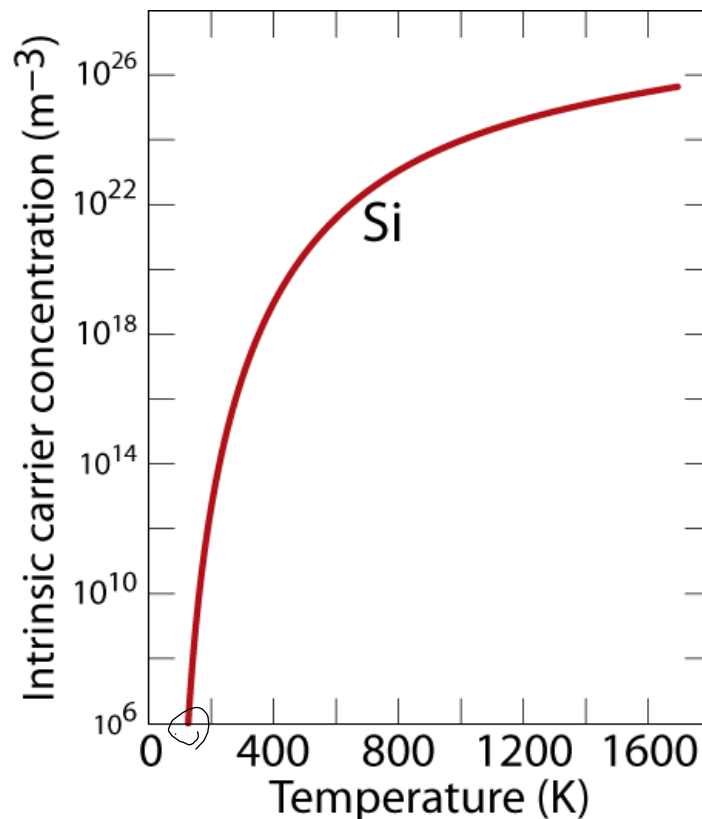
$$n_i = \frac{\sigma}{|e|(\mu_e + \mu_h)} = \frac{10^{-6}(\Omega \cdot \text{m})^{-1}}{(1.6 \times 10^{-19} \text{ C})(0.85 + 0.45 \text{ m}^2/\text{V} \cdot \text{s})}$$

$$\text{For GaAs} \quad n_i = 7 \times 10^{12} \text{ m}^{-3}$$

$$\text{For Si} \quad n_i = 1.3 \times 10^{16} \text{ m}^{-3}$$

Intrinsic Semiconductors: Conductivity vs T

- Data for **Pure Silicon**:
 - σ increases with T
 - opposite to metals



Adapted from Fig. 18.16,
Callister & Rethwisch 8e.

$$\sigma = n_i |e| (\mu_e + \mu_h)$$

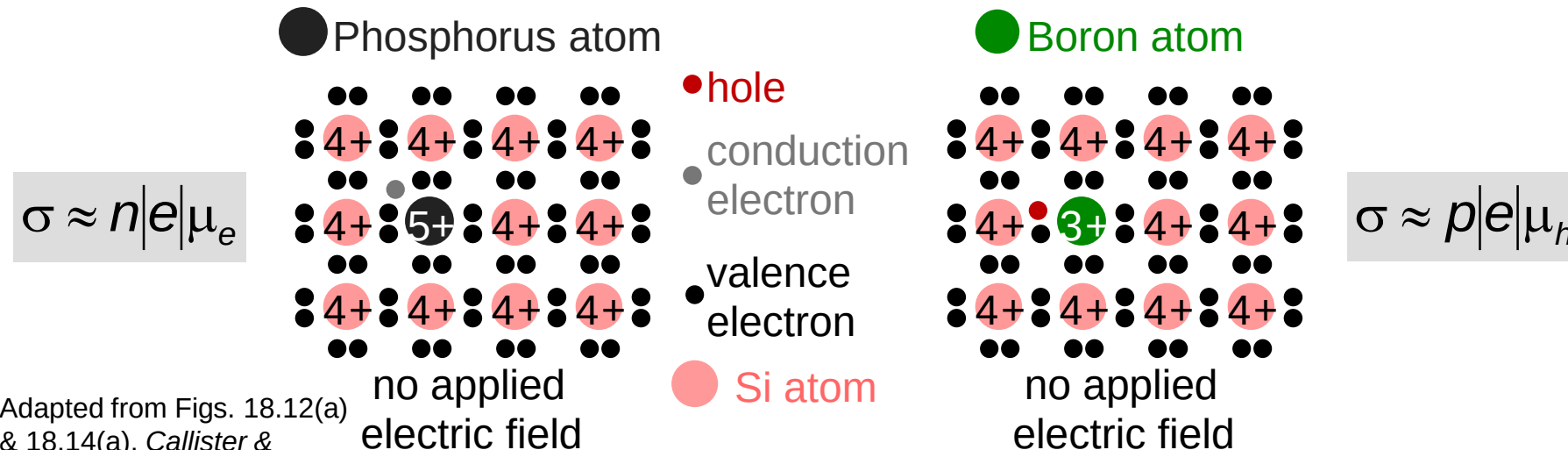
$$n_i \propto e^{-E_{\text{gap}} / kT}$$

material	band gap (eV)
Si	1.11
Ge	0.67
GaP	2.25
CdS	2.40

Selected values from Table 18.3,
Callister & Rethwisch 8e.

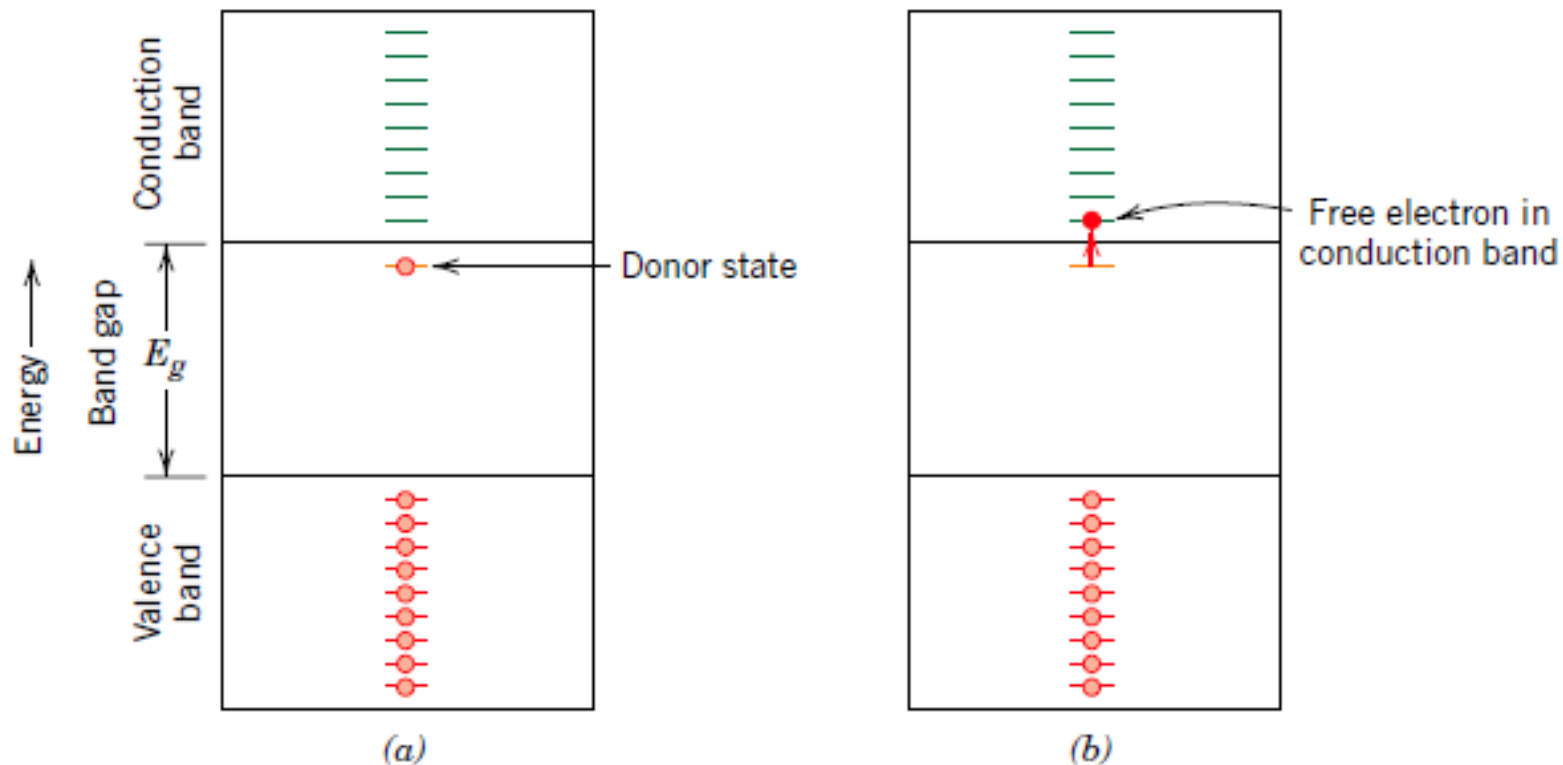
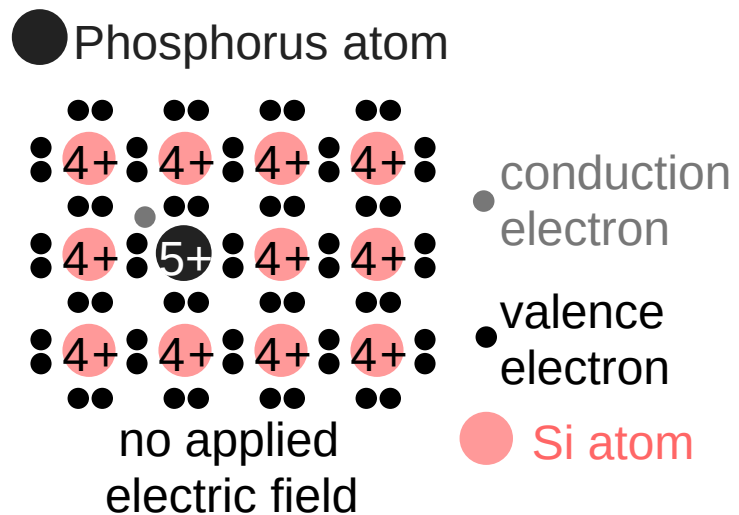
Intrinsic vs Extrinsic Conduction

- **Intrinsic:**
 - case for pure Si
 - # electrons = # holes ($n = p$)
- **Extrinsic:**
 - electrical behavior is determined by presence of impurities that introduce excess electrons or holes
 - $n \neq p$
- **n -type Extrinsic:** ($n \gg p$)
- **p -type Extrinsic:** ($p \gg n$)



Adapted from Figs. 18.12(a) & 18.14(a), Callister & Rethwisch 8e.

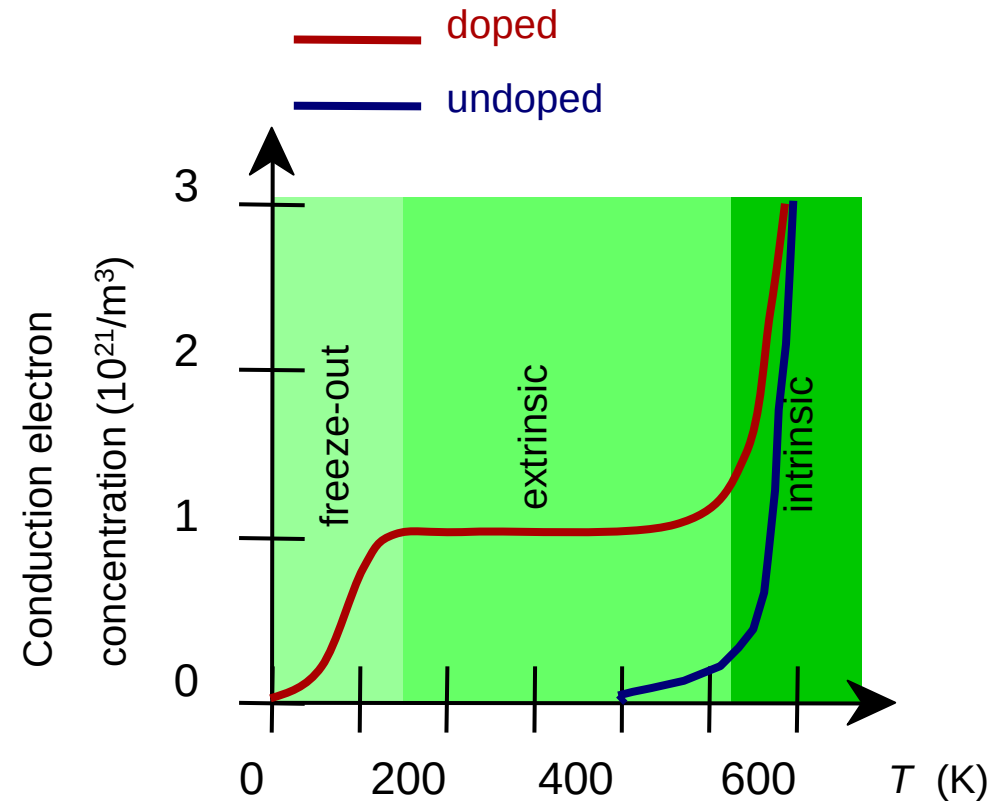
- n*-type** Extrinsic: ($n \gg p$)



Will Zn act as a donor or acceptor when added to the compound semiconductor GaAs? Why? (Assume that Zn is a substitutional impurity.)

Extrinsic Semiconductors: Conductivity vs. Temperature

- Data for **Doped Silicon**:
 - σ increases doping
 - reason: imperfection sites lower the activation energy to produce mobile electrons.
- Comparison: **intrinsic** vs **extrinsic** conduction...
 - extrinsic doping level: $10^{21}/\text{m}^3$ of a *n*-type donor impurity (such as P).
 - for $T < 100$ K: "freeze-out", thermal energy insufficient to excite electrons.
 - for $150 \text{ K} < T < 450 \text{ K}$: "extrinsic"
 - for $T \gg 450 \text{ K}$: "intrinsic"



Adapted from Fig. 18.17, *Callister & Rethwisch 8e*. (Fig. 18.17 from S.M. Sze, *Semiconductor Devices, Physics, and Technology*, Bell Telephone Laboratories, Inc., 1985.)

Electrical Conductivity Determination for Intrinsic Silicon at 150°C

Calculate the electrical conductivity of intrinsic silicon at 150° C (423 K).

$$\sigma = n|e|\mu_e + p|e|\mu_h, \quad n = p = n_i$$

$$\begin{aligned}\sigma &= \underline{n_i} |e| (\mu_e + \mu_h), \quad n_i \propto \exp\left(\frac{-E_g}{KT}\right) \\ &= 4 \times 10^{19} [\text{m}^{-3}] (1.6 \times 10^{-19}) \left(0.06 \frac{\text{m}^2}{\text{V}\cdot\text{s}} + 0.022 \frac{\text{m}^2}{\text{V}\cdot\text{s}}\right) \\ &= 0.52 (\Omega \cdot \text{m})^{-1}\end{aligned}$$

Room-Temperature and Elevated-Temperature Electrical Conductivity Calculations for Extrinsic Silicon

To high-purity silicon is added 10^{23} m^{-3} arsenic atoms.

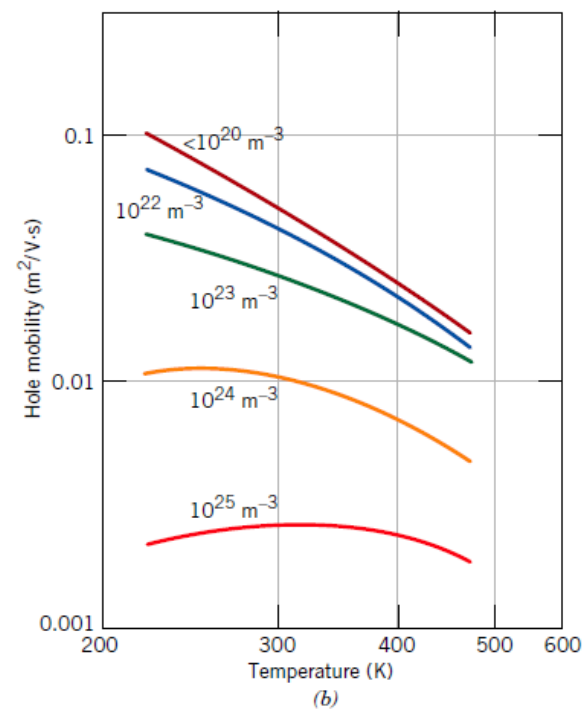
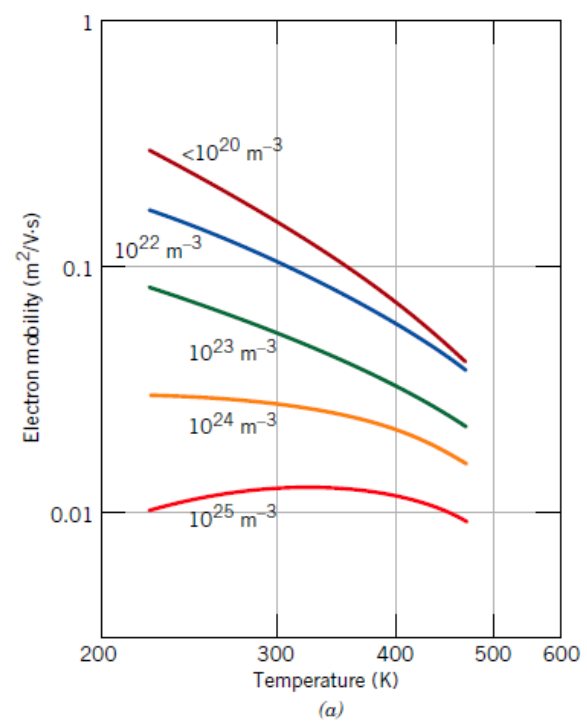
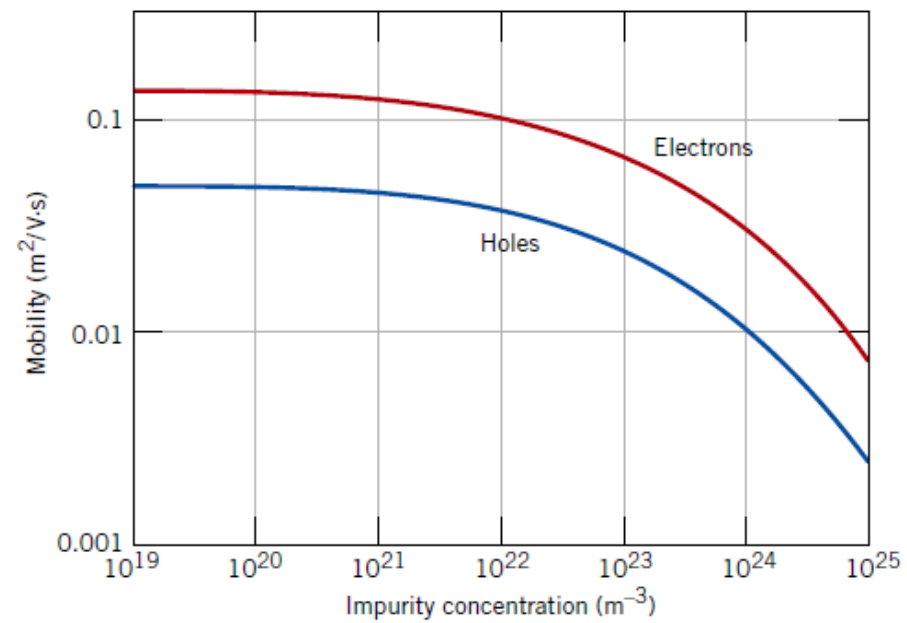
- (a) Is this material *n*-type or *p*-type?
- (b) Calculate the room-temperature electrical conductivity of this material.
- (c) Compute the conductivity at 100°C (373 K).

a) *n*-type

b) $\sigma = n |e| \mu_e = (10^{23} \text{ m}^{-3}) (1.6 \times 10^{-19} \text{ C}) (0.07) \frac{\text{m}^2}{\text{V}\cdot\text{s}}$
 $= 1120 (\Omega \cdot \text{m})^{-1}$

c) $\sigma = n |e| \mu_e$

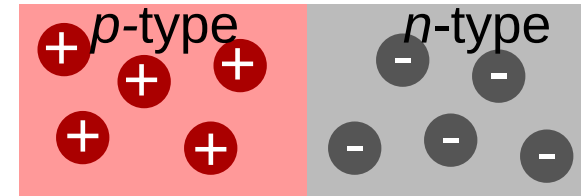
$$= (10^{23} \text{ m}^{-3}) (1.6 \times 10^{-19} \text{ C}) (0.09) \frac{\text{m}^2}{\text{V}\cdot\text{s}}$$
$$= 1440 (\Omega \cdot \text{m})^{-1}$$



p - n Rectifying Junction

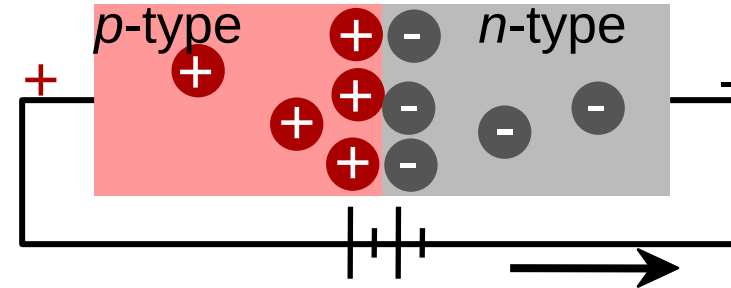
- Allows flow of electrons in one direction only (e.g., useful to convert alternating current to direct current).
- Processing: diffuse P into one side of a B-doped crystal.

-- No applied potential:
no net current flow.

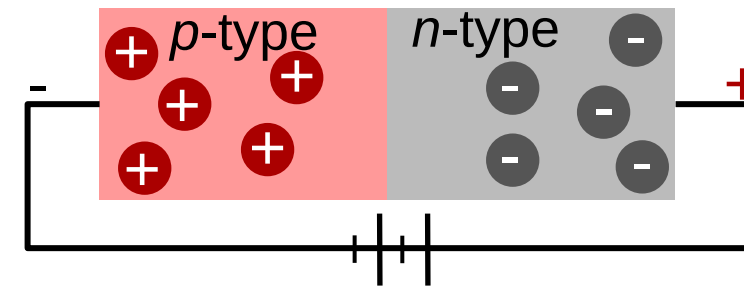


Adapted from
Fig. 18.21
Callister &
Rethwisch
8e.

-- Forward bias: carriers
flow through p -type and
 n -type regions; holes and
electrons recombine at
 p - n junction; current flows.



-- Reverse bias: carriers
flow away from p - n junction;
junction region depleted of
carriers; little current flow.



Properties of Rectifying Junction

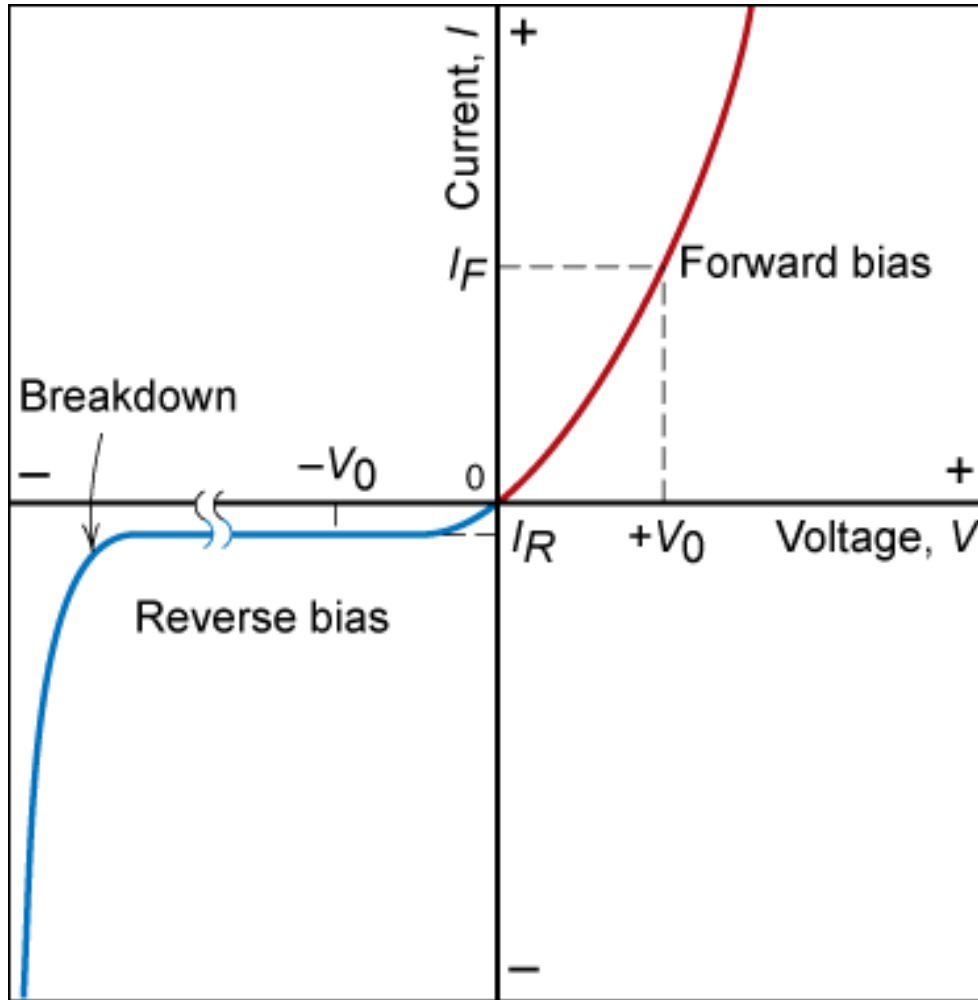


Fig. 18.22, Callister & Rethwisch 8e.

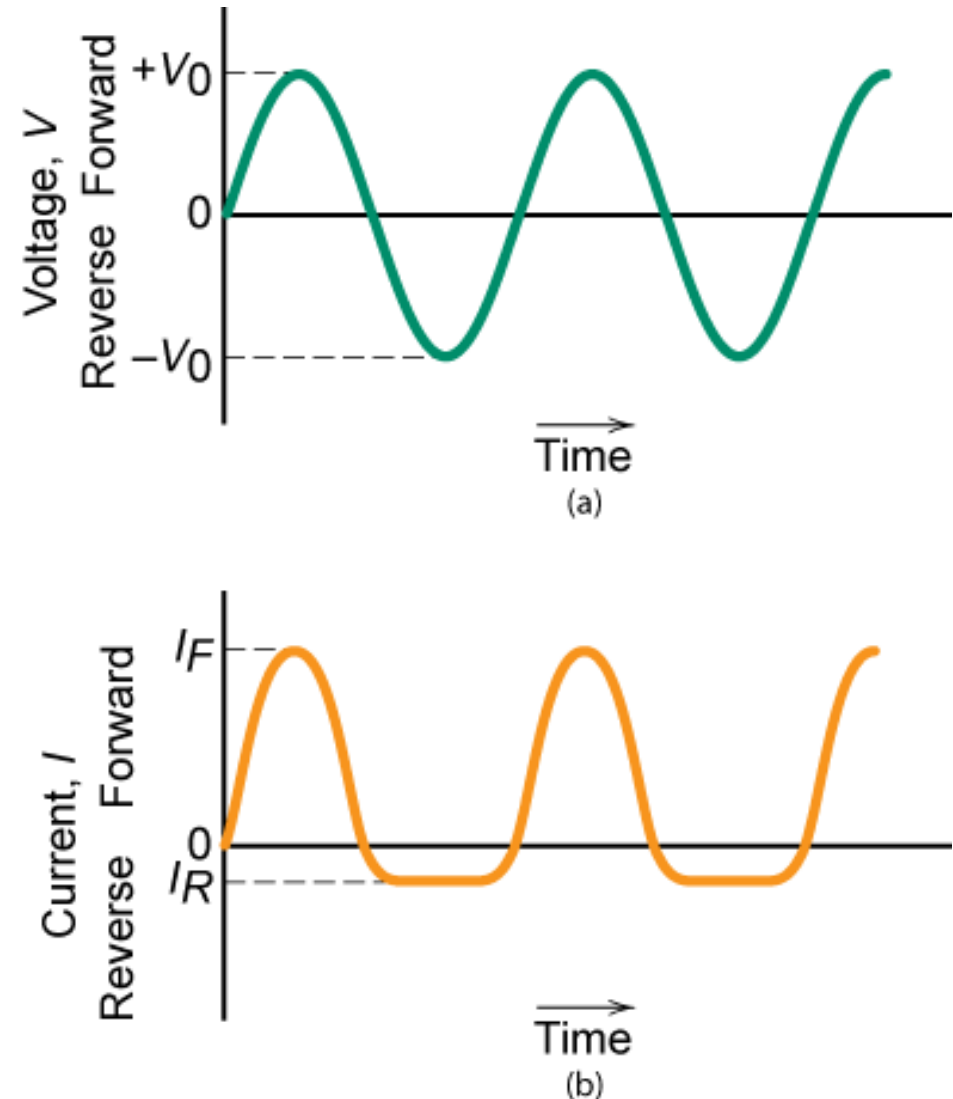


Fig. 18.23, Callister & Rethwisch 8e.

MOSFET Transistor

Integrated Circuit Device

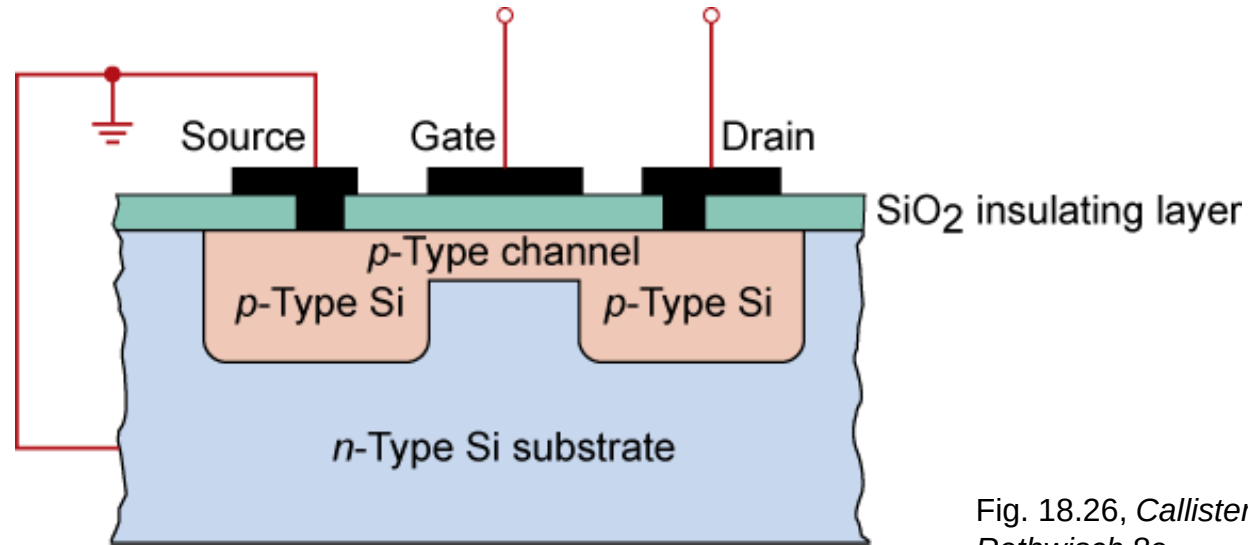
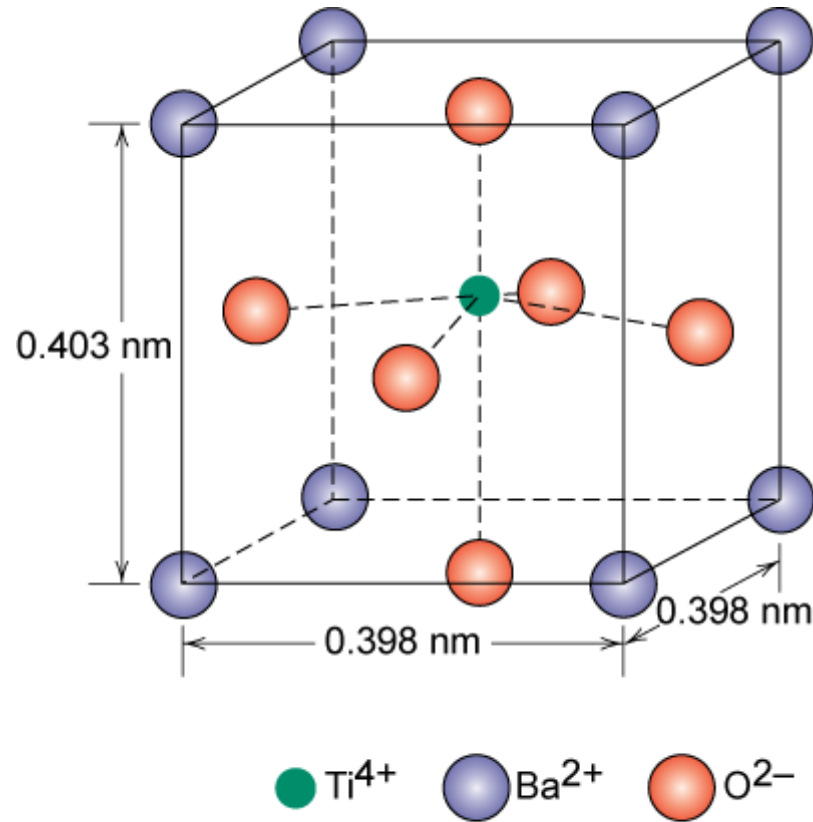


Fig. 18.26, Callister & Rethwisch 8e.

- MOSFET (metal oxide semiconductor field effect transistor)
- Integrated circuits - state of the art ca. 50 nm line width
 - ~ 1,000,000,000 components on chip
 - chips formed one layer at a time

Ferroelectric Ceramics

- Experience spontaneous polarization



(a)

BaTiO₃ -- ferroelectric below its Curie temperature (120°C)

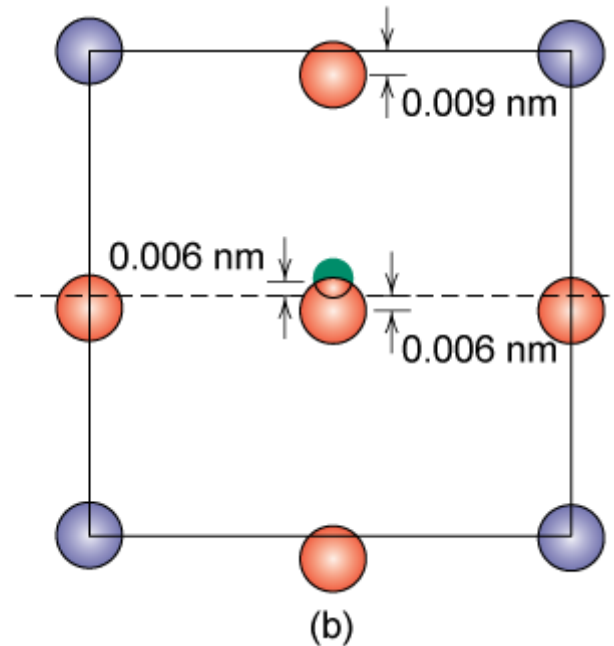
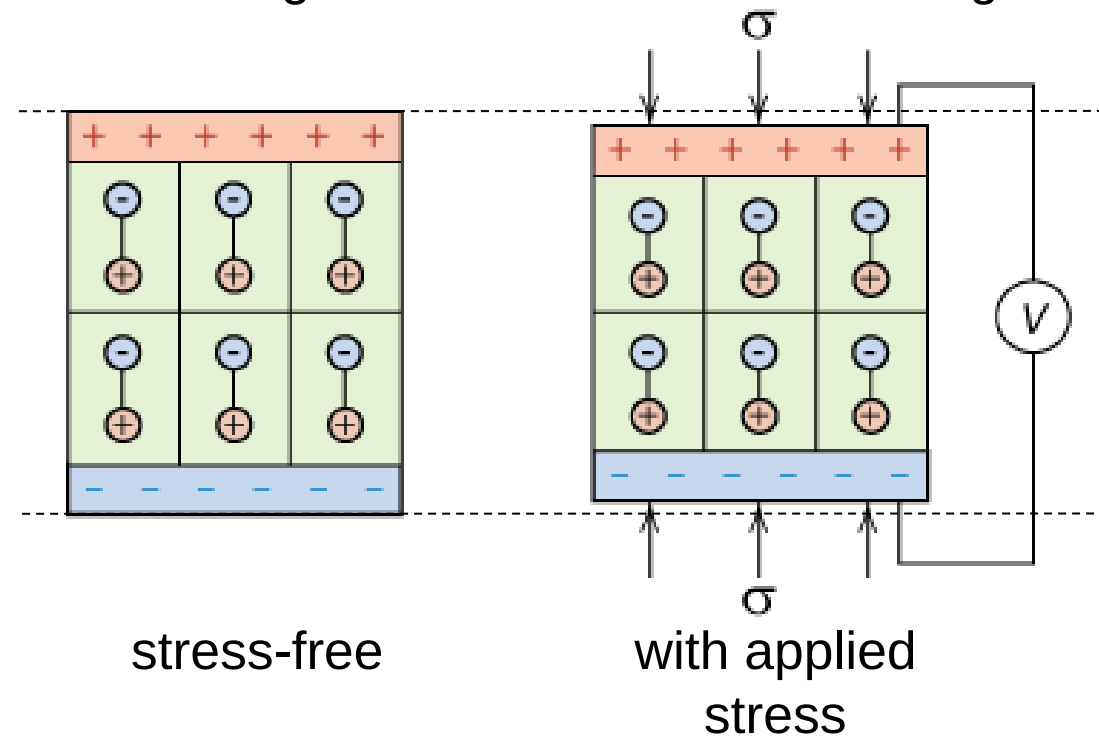


Fig. 18.35, Callister & Rethwisch 8e.

Piezoelectric Materials

Piezoelectricity

- application of stress induces voltage
- application of voltage induces dimensional change



Adapted from Fig. 18.36, *Callister & Rethwisch 8e*. (Fig. 18.36 from Van Vlack, Lawrence H., *Elements of Materials Science and Engineering*, 1989, p.482, Adapted by permission of Pearson Education, Inc., Upper Saddle River, New Jersey.)

Summary

- Electrical *conductivity* and *resistivity* are:
 - material parameters
 - geometry independent
- Conductors, semiconductors, and insulators...
 - differ in range of conductivity values
 - differ in availability of electron excitation states
- For metals, *resistivity* is increased by
 - increasing temperature
 - addition of imperfections
 - plastic deformation
- For pure semiconductors, *conductivity* is increased by
 - increasing temperature
 - doping [e.g., adding B to Si (*p*-type) or P to Si (*n*-type)]
- Other electrical characteristics
 - ferroelectricity
 - piezoelectricity

- 18.11 At room temperature the electrical conductivity and the electron mobility for copper are $6.0 \times 10^7 (\Omega \cdot \text{m})^{-1}$ and $0.0030 \text{ m}^2/\text{V} \cdot \text{s}$, respectively. (a) Compute the number of free electrons per cubic meter for copper at room temperature. (b) What is the number of free electrons per copper atom? Assume a density of 8.9 g/cm^3 .
- 18.21 At room temperature the electrical conductivity of PbTe is $550 (\Omega \cdot \text{m})^{-1}$, whereas the electron and hole mobilities are 0.16 and $0.080 \text{ m}^2/\text{V} \cdot \text{s}$, respectively. Compute the intrinsic carrier concentration for PbTe at room temperature.
- 18.25 An *n*-type semiconductor is known to have an electron concentration of $3 \times 10^{18} \text{ m}^{-3}$. If the electron drift velocity is 100 m/s in an electric field of 600 V/m , calculate the conductivity of this material.
- 18.29 (a) The room-temperature electrical conductivity of a silicon specimen is $5.93 \times 10^{-3} (\Omega \cdot \text{m})^{-1}$. The hole concentration is known to be $7.0 \times 10^{17} \text{ m}^{-3}$. Using the electron and hole mobilities for silicon in Table 18.3, compute the electron concentration. (b) On the basis of the result in part (a), is the specimen intrinsic, *n*-type extrinsic, or *p*-type extrinsic? Why?